

BEE 4750 Homework 3: Dissolved Oxygen and Monte Carlo

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Due Date

Thursday, 10/16/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to implement a model for dissolved oxygen in a river with multiple waste releases and use this to develop a strategy to ensure regulatory compliance.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `~/hw3-claire`

```
using Random
using Plots
using LaTeXStrings
using Distributions
```

Problems (Total: 30 Points)

Problem 1 (30 points)

A river which flows at 6 km/d is receiving waste discharges from two sources which are 15 km apart. The oxygen reaeration rate is 0.55 day⁻¹, and the decay rates of CBOD and NBOD are 0.35 and 0.25 day⁻¹, respectively. The river's saturated dissolved oxygen concentration is 10 mg/L.

If the characteristics of the river inflow and waste discharges are given in Table 1, write a Julia model to compute the dissolved oxygen concentration from the first wastewater discharge to an arbitrary distance d km downstream. Use your model to compute the minimum dissolved oxygen concentration up to 50 km downstream and how far downriver this maximum occurs.

Parameter	River Inflow	Waste Stream 1	Waste Stream 2
Inflow	100,000 m ³ /d	10,000 m ³ /d	15,000 m ³ /d
DO Concentration	7.5 mg/L	5 mg/L	5 mg/L
CBOD	5 mg/L	50 mg/L	45 mg/L
NBOD	5 mg/L	35 mg/L	35 mg/L

Table 1: River inflow and waste stream characteristics for Problem 1.

Problem 1.1

Implement the Streeter-Phelps (analytic) solution for the dissolved oxygen concentration. Plot the dissolved oxygen concentration from the first waste stream to 50 km downriver. What is the minimum value in mg/L?

```
x2 = 15 # km
xf = 50 # km
dx = 0.1 # km
u = 6 # km/d

C_sat = 10 # mg/L

kr = 0.55 # day-1
kc = 0.35 # day-1
kn = 0.25 # day-1

Qr = 100000 # m3/day
Cdo_r = 7.5 # mg/L
```

```

Cc_r = 5 # mg/L
Cn_r = 5 # mg/L

Q1 = 10000 # m^3/day
Cdo_1 = 5 # mg/L
Cc_1 = 50 # mg/L
Cn_1 = 35 # mg/L

Q2 = 15000 # m^3/day
Cdo_2 = 5 # mg/L
Cc_2 = 45 # mg/L
Cn_2 = 35 # mg/L

C_new(q1, c1, q2, c2) = (q1 * c1 + q2 * c2) / (q1 + q2)

function mix(DO_0, CBOD0, NBOD0, t, k1, k2, k3, sat)
    DO_new = DO_0 * exp(-k1 * t)
    CBOD_new = CBOD0 * (k2 / (k1 - k2)) * (exp(-k2 * t) - exp(-k1 * t))
    NBOD_new = NBOD0 * (k3 / (k1 - k3)) * (exp(-k3 * t) - exp(-k1 * t))
    D = DO_new + CBOD_new + NBOD_new
    return sat - D
end

Q_first = Qr + Q1
DO_mix1 = C_new(Qr, Cdo_r, Q1, Cdo_1)
CBOD_mix1 = C_new(Qr, Cc_r, Q1, Cc_1)
NBOD_mix1 = C_new(Qr, Cn_r, Q1, Cn_1)
DO_first = C_sat - DO_mix1

decay(C_0, k, t) = C_0 * exp(-k * t)

xs = 0.0:dx:xf

for (i, x) in enumerate(xs)
    if x <= x2
        t = x / u
        DOs[i] = mix(DO_first, CBOD_mix1, NBOD_mix1, t, kr, kc, kn, C_sat)
    else
        t_x2 = x2 / u
        DO_x2 = mix(DO_first, CBOD_mix1, NBOD_mix1, t_x2, kr, kc, kn, C_sat)
        CBOD_x2 = decay(CBOD_mix1, kc, t_x2)
        NBOD_x2 = decay(NBOD_mix1, kn, t_x2)
    end
end

```

```

        DO_mix2 = C_new(Q_first, DO_x2, Q2, Cdo_2)
        CBOD_mix2 = C_new(Q_first, CBOD_x2, Q2, Cc_2)
        NBOD_mix2 = C_new(Q_first, NBOD_x2, Q2, Cn_2)
        DO_second = C_sat - DO_mix2
        t_after2 = (x - x2) / u
        DOs[i] = mix(DO_second, CBOD_mix2, NBOD_mix2, t_after2, kr, kc, kn, C_sat)
    end
end

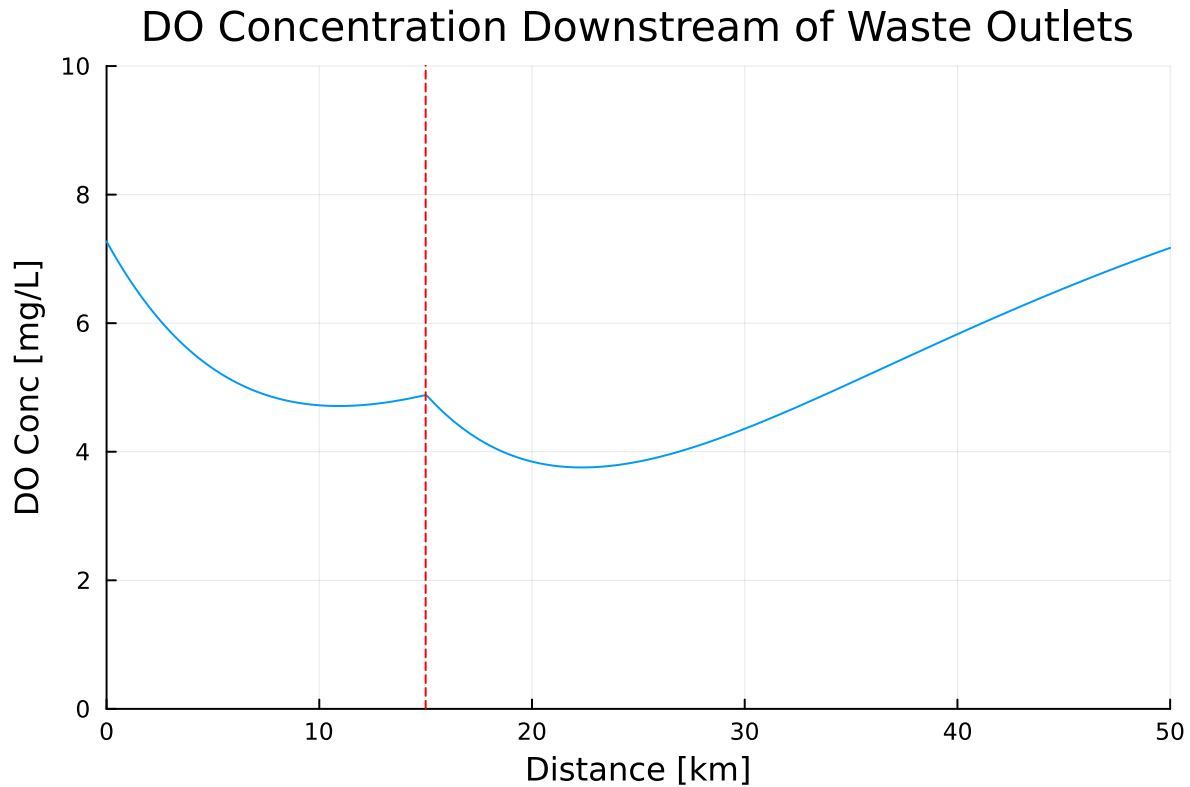
min_conc=minimum(DOs)

p = plot(xs, DOs,
    xlim = (0, 50),
    ylim = (0,10),
    legend = false,
    xlabel = "Distance [km]",
    ylabel = "DO Conc [mg/L]",
    title = "DO Concentration Downstream of Waste Outlets"
)

vline!(p, [15], color=:red, linestyle=:dash)

display(p)
println("Minimum dissolved oxygen concentration = ", min_conc)

```



Minimum dissolved oxygen concentration = 3.75570046982907

Problem 1.2

Implement a numerically-integrated (discretized) solution for the dissolved oxygen concentration. Using a resolution of 0.5km, conduct a simulation and plot the results on the same axis as the plot from Problem 1.1. How has the minimum value changed? What do you attribute this difference to (be specific about the source of the difference(s) in terms of the simulation dynamics, not just that one is a numerical approximation).

The new minimum is ~0.9 mg/L higher than in the analytic solution. This can likely be attributed to some truncation error from the larger time step as compared to the analytical solution and the fact that the values are only incrementally evaluated. Also, the numerical integration method assumes constant values and only has values compared to those 0.5 km before, ignoring what goes on in between those time steps meaning a minimum will be mostly ignored if it lies between two time steps. This means a change could be ignored even further as the effects of evaluating based on that point will also be ignored.

```

dx_int = 0.5 # km
xs_int = 0.0:dx_int:xf
dt_int = dx_int / u

DO_int=zeros(length(xs_int))
CBOD_int = zeros(length(xs_int))
NBOD_int = zeros(length(xs_int))

DO_int[1] = DO_mix1
D=C_sat-DO_mix1
CBOD_int[1] = CBOD_mix1
NBOD_int[1] = NBOD_mix1

for i in 2:length(xs_int)
    CBOD_int[i] = CBOD_int[i-1] * exp(-kc*dt_int)
    NBOD_int[i] = NBOD_int[i-1] * exp(-kn*dt_int)

    D = D * exp(-kr*dt_int) + (CBOD_int[i-1] - CBOD_int[i])+(NBOD_int[i-1]-NBOD_int[i])

    DO_int[i] = C_sat - D

    if xs_int[i-1] < x2 < xs_int[i]
        DO_int[i] = C_new(Q_first, DO_int[i], Q2, Cdo_2)
        CBOD_int[i] = C_new(Q_first, CBOD_int[i], Q2, Cc_2)
        NBOD_int[i] = C_new(Q_first, NBOD_int[i], Q2, Cn_2)
        DO= C_sat - DO_int[i]
        Q_first += Q2
    end
end

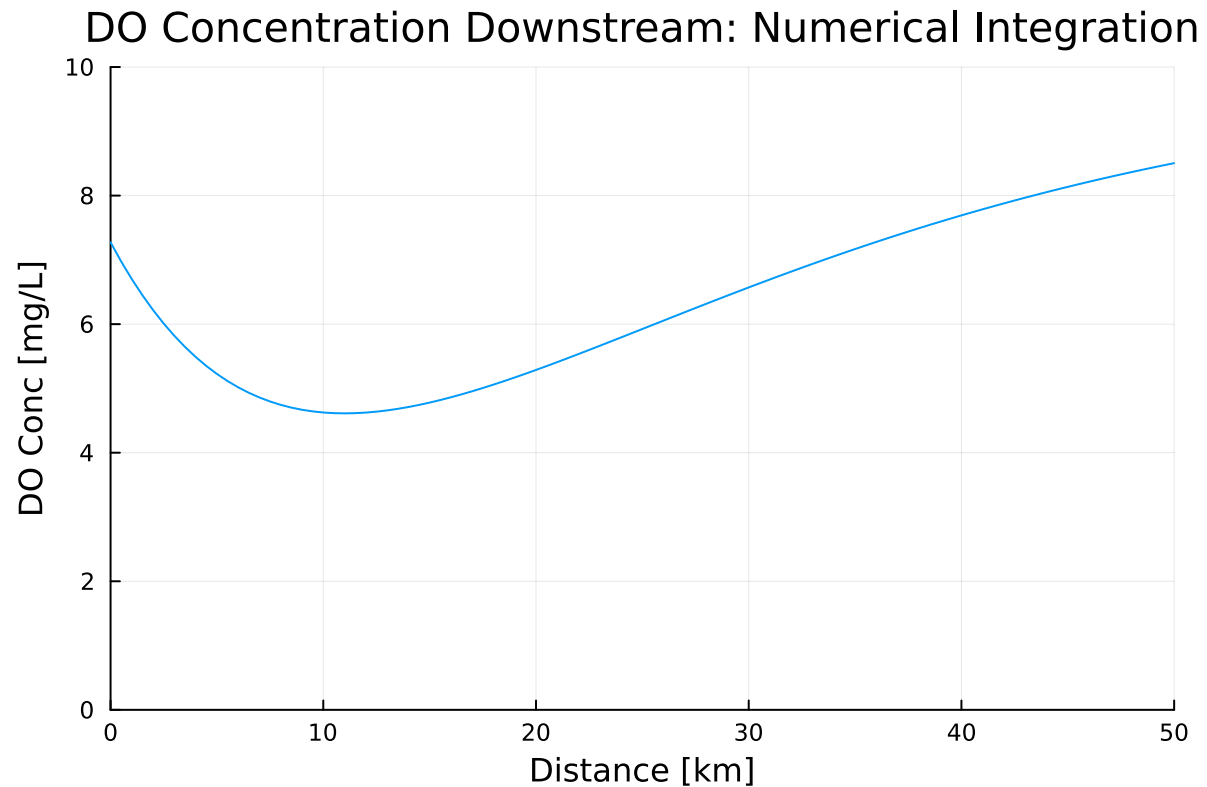
min_conc_new=minimum(DO_int)

p2 = plot(xs_int, DO_int,
    xlim = (0, 50),
    ylim = (0,10),
    legend = false,
    xlabel = "Distance [km]",
    ylabel = "DO Conc [mg/L]",
    title = "DO Concentration Downstream: Numerical Integration"
)

display(p2)

```

```
println("Minimum dissolved oxygen concentration = ", min_conc_new)
```



Minimum dissolved oxygen concentration = 4.612126754365276

Problem 1.3

Using the analytic model, what is the minimum level of treatment (% removal of organic waste; assume this is equivalent to the same level of reduction of the CBOD and NBOD) for waste stream 1 that will ensure that the dissolved oxygen concentration is in compliance with the 4 mg/L standard along with a 5% margin of safety, assuming that waste stream 2 remains untreated? How about if only waste stream 2 is treated?

```
perc = 0.29
treated = 1-perc

Cdo_1_treated = 5*treated # mg/L
Cc_1_treated = 50*treated # mg/L
Cn_1_treated = 35*treated # mg/L
```

```

Q_first = Qr + Q1
DO_mix1 = C_new(Qr, Cdo_r, Q1, Cdo_1_treated)
CBOD_mix1 = C_new(Qr, Cc_r, Q1, Cc_1_treated)
NBOD_mix1 = C_new(Qr, Cn_r, Q1, Cn_1_treated)
DO_first = C_sat - DO_mix1

decay(C_0, k, t) = C_0 * exp(-k * t)

xs = 0.0:dx:xf

for (i, x) in enumerate(xs)
    if x <= x2
        t = x / u
        DOs[i] = mix(DO_first, CBOD_mix1, NBOD_mix1, t, kr, kc, kn, C_sat)
    else
        t_x2 = x2 / u
        DO_x2 = mix(DO_first, CBOD_mix1, NBOD_mix1, t_x2, kr, kc, kn, C_sat)
        CBOD_x2 = decay(CBOD_mix1, kc, t_x2)
        NBOD_x2 = decay(NBOD_mix1, kn, t_x2)
        DO_mix2 = C_new(Q_first, DO_x2, Q2, Cdo_2)
        CBOD_mix2 = C_new(Q_first, CBOD_x2, Q2, Cc_2)
        NBOD_mix2 = C_new(Q_first, NBOD_x2, Q2, Cn_2)
        DO_second = C_sat - DO_mix2
        t_after2 = (x - x2) / u
        DOs[i] = mix(DO_second, CBOD_mix2, NBOD_mix2, t_after2, kr, kc, kn, C_sat)
    end
end

min_conc=minimum(DOs)

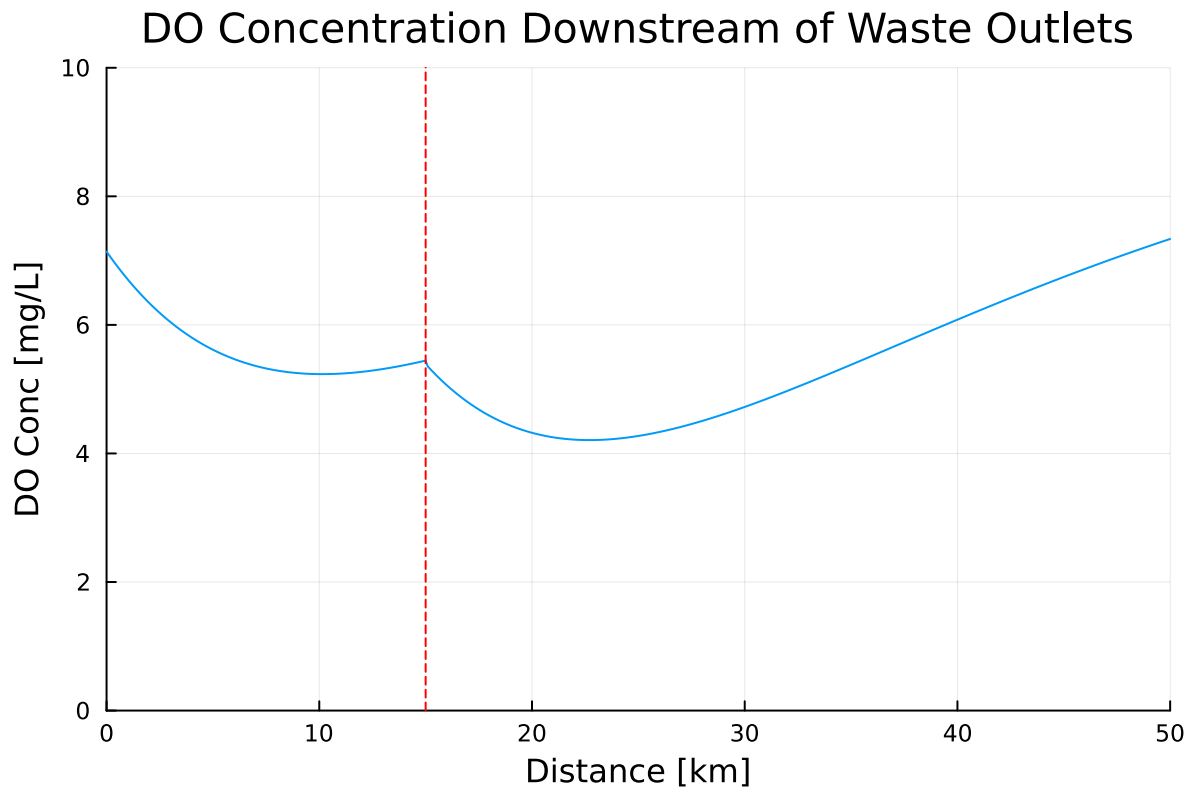
p = plot(xs, DOs,
    xlim = (0, 50),
    ylim = (0,10),
    legend = false,
    xlabel = "Distance [km]",
    ylabel = "DO Conc [mg/L]",
    title = "DO Concentration Downstream of Waste Outlets"
)

vline!(p, [15], color=:red, linestyle=:dash)

display(p)

```

```
println("Minimum dissolved oxygen concentration = ", min_conc)
```



Minimum dissolved oxygen concentration = 4.20792327031352

Problem 1.4

Suppose you are responsible for designing a waste treatment plan for discharges into the river, with a regulatory mandate to keep the dissolved oxygen concentration above 4 mg/L. Discuss whether you'd opt to treat waste stream 2 alone or both waste streams equally. What other information might you need to make a conclusion, if any?

Problem 1.5

Suppose that it is known that the DO concentrations at the river inflow can vary according to a LogNormal(2.0, 0.15) distribution. Conduct a Monte Carlo simulation of the DO concentration in the river. If you treat only waste stream 1 (based on your analysis from Problem 1.3), what is the expected probability and 95% confidence interval that the river fails to comply with the

regulatory standard of 4 mg/L? How did you decide that your Monte Carlo sample size was sufficiently large?

References

List any external references consulted, including classmates.